Layered repositories, such as an archive containing copies of a subset of records from another archive or archives that have moved or been bought by other archives, are not modeled in this version of the specification. It is expected they will be added in a later version. Until such time, it is recommended that the repository record store current contact information, if known.

SHARED_NOTE_RECORD :=

```
n @XREF:SNOTE@ SNOTE <Text>                    {1:1}  g7:record-SNOTE
  +1 MIME <MediaType>                          {0:1}  g7:MIME
  +1 LANG <Language>                           {0:1}  g7:LANG
  +1 TRAN <Text>                               {0:M}  g7:NOTE-TRAN
     +2 MIME <MediaType>                       {0:1}  g7:MIME
     +2 LANG <Language>                        {0:1}  g7:LANG
  +1 <<SOURCE_CITATION>>                       {0:M}
  +1 <<IDENTIFIER_STRUCTURE>>                  {0:M}
  +1 <<CHANGE_DATE>>                           {0:1}
  +1 <<CREATION_DATE>>                         {0:1}
```

A catch-all location for information that does not fully fit within other structures. It may include research notes, additional context, alternative interpretations, reasoning, and so forth.

A shared note record may be pointed to by multiple other structures. Shared notes should only be used if editing the note in one place should edit it in all other places or if the note itself requires an `IDENTIFIER_STRUCTURE` (p.48). If each instance of the note may be edited separately and no identifier is needed, a `NOTE` (p.80) should be used instead.

Each `SNOTE.TRAN` (p.89) must have either a `MIME` (p.78) or `LANG` (p.75) substructure or both.

> *Example* — The origin of a name might be a reasonable shared note, while the reason a particular person was given that name may make more sense as a non-shared note.
>
> ```
> 0 @GORDON@ SNOTE "Gordon" is a traditional Scottish surname.
> 1 CONT It became a given name in honor of Charles George Gordon.
> 0 @I1@ INDI
> 1 NAME Gordon /Jones/
> 2 NOTE Named after the astronaut Gordon Cooper
> 2 SNOTE @GORDON@
> ```

> *Note* — The ability to have multiple structures share a single note using pointers was introduced in version 5.0 in 1991. However, as of 2021 relatively few applications have a user interface that presents shared notes as such to users. It is recommended that `SNOTE` (p.85) be avoided when `NOTE` will suffice.

A `SHARED_NOTE_RECORD` (p.41) may contain a pointer to a `SOURCE_RECORD` and vice versa. Applications must not create datasets where these mutual pointers form a cycle. Applications should also ensure they can handle invalid files with such cycles in a safe manner.

SOURCE_RECORD :=

```
n @XREF:SOUR@ SOUR                          {1:1}  g7:record-SOUR
  +1 DATA                                   {0:1}  g7:DATA
     +2 EVEN <List:Enum>                    {0:M}  g7:DATA-EVEN
        +3 DATE <DatePeriod>                {0:1}  g7:DATA-EVEN-DATE
           +4 PHRASE <Text>                 {0:1}  g7:PHRASE
        +3 <<PLACE_STRUCTURE>>              {0:1}
     +2 AGNC <Text>                         {0:1}  g7:AGNC
     +2 <<NOTE_STRUCTURE>>                  {0:M}
  +1 AUTH <Text>                            {0:1}  g7:AUTH
  +1 TITL <Text>                            {0:1}  g7:TITL
  +1 ABBR <Text>                            {0:1}  g7:ABBR
  +1 PUBL <Text>                            {0:1}  g7:PUBL
  +1 TEXT <Text>                            {0:1}  g7:TEXT
     +2 MIME <MediaType>                    {0:1}  g7:MIME
     +2 LANG <Language>                     {0:1}  g7:LANG
  +1 <<SOURCE_REPOSITORY_CITATION>>         {0:M}
  +1 <<IDENTIFIER_STRUCTURE>>               {0:M}
  +1 <<NOTE_STRUCTURE>>                     {0:M}
  +1 <<MULTIMEDIA_LINK>>                    {0:M}
  +1 <<CHANGE_DATE>>                        {0:1}
  +1 <<CREATION_DATE>>                      {0:1}
```

A source record describes an entire source. A source may also point to `REPO` (p.83)s to describe repositories or archives where the source document may be found. The part of a source relevant to a specific fact, such as a specific page or entry, is indicated in a `SOURCE_CITATION` (p.58) that points to the source record.

> *Note* — This sourcing model is known to be insufficient for some use cases and may be refined in a future version of this specification.

A `SOURCE_RECORD` (p.42) may contain a pointer to a `SHARED_NOTE_RECORD` (p.41) and vice versa. Applications must not create datasets where these mutual pointers form a cycle. Applications should also ensure they can handle invalid files with such cycles in a safe manner.

SUBMITTER_RECORD :=

```
n @XREF:SUBM@ SUBM                          {1:1}  g7:record-SUBM
  +1 NAME <Text>                            {1:1}  g7:NAME
  +1 <<ADDRESS_STRUCTURE>>                  {0:1}
  +1 PHON <Special>                         {0:M}  g7:PHON
  +1 EMAIL <Special>                        {0:M}  g7:EMAIL
  +1 FAX <Special>                          {0:M}  g7:FAX
  +1 WWW <Special>                          {0:M}  g7:WWW
  +1 <<MULTIMEDIA_LINK>>                    {0:M}
  +1 LANG <Language>                        {0:M}  g7:SUBM-LANG
  +1 <<IDENTIFIER_STRUCTURE>>               {0:M}
  +1 <<NOTE_STRUCTURE>>                     {0:M}
  +1 <<CHANGE_DATE>>                        {0:1}
  +1 <<CREATION_DATE>>                      {0:1}
```

The submitter record identifies an individual or organization that contributed information contained in the dataset. All records in the document are assumed to be contributed by the submitter referenced in the `HEAD` (p.74), unless a `SUBM` (p.86) structure inside a specific record points at a different submitter record.

3.2.3. Substructures

ADDRESS_STRUCTURE :=

```
n ADDR <Special>                            {1:1}  g7:ADDR
  +1 ADR1 <Special>                         {0:1}  g7:ADR1
  +1 ADR2 <Special>                         {0:1}  g7:ADR2
  +1 ADR3 <Special>                         {0:1}  g7:ADR3
  +1 CITY <Special>                         {0:1}  g7:CITY
  +1 STAE <Special>                         {0:1}  g7:STAE
  +1 POST <Special>                         {0:1}  g7:POST
  +1 CTRY <Special>                         {0:1}  g7:CTRY
```

A specific building, plot, or location. The payload is the full formatted address as it would appear on a mailing label, including appropriate line breaks (encoded using `CONT` (p.68) tags). The expected order of address components varies by region; the address should be organized as expected by the addressed region.

Optionally, additional substructures such as `STAE` (p.86) and `CTRY` (p.69) are provided to be used by systems that have structured their addresses for indexing and sorting. If the substructures and `ADDR` (p.65) payload disagree, the `ADDR` payload shall be taken as correct. Because the regionally-correct order and formatting of address components cannot be determined from the substructures alone, the `ADDR` payload is required, even if its content appears to be redundant with the substructures.

> *Deprecation Note* — ADR1 (p.65) and ADR2 (p.65) were introduced in version 5.5 (1996) and ADR3 (p.66) in version 5.5.1 (1999), defined as "The first/second/third line of an address." Some applications interpreted ADR1 as "the first line of the *street* address", but most took the spec as-written and treated it as a straight copy of a line of text already available in the ADDR payload.
>
> Duplicating information bloats files and introduces the potential for self-contradiction. ADR1, ADR2, and ADR3 should not be added to new files.

ASSOCIATION_STRUCTURE :=

```
n ASSO @<XREF:INDI>@                          {1:1}  g7:ASSO
  +1 PHRASE <Text>                            {0:1}  g7:PHRASE
  +1 ROLE <Enum>                              {1:1}  g7:ROLE
     +2 PHRASE <Text>                         {0:1}  g7:PHRASE
  +1 <<NOTE_STRUCTURE>>                       {0:M}
  +1 <<SOURCE_CITATION>>                      {0:M}
```

An individual associated with the subject of the superstructure. The nature of the association is indicated in the ROLE (p.84) substructure.

A voidPtr (p.11) and PHRASE (p.81) can be used to describe associations to people not referenced by any INDI record.

> *Example* — The following indicates that "Mr Stockdale" was the individual's teacher and that individual @I2@ was the clergy officiating at their baptism.
>
> ```
> 0 @I1@ INDI
> 1 ASSO @VOID@
> 2 PHRASE Mr Stockdale
> 2 ROLE OTHER
> 3 PHRASE Teacher
> 1 BAPM
> 2 DATE 1930
> 2 ASSO @I2@
> 3 ROLE CLERGY
> ```

CHANGE_DATE :=

```
n CHAN                                        {1:1}  g7:CHAN
  +1 DATE <DateExact>                         {1:1}  g7:DATE-exact
     +2 TIME <Time>                           {0:1}  g7:TIME
  +1 <<NOTE_STRUCTURE>>                       {0:M}
```

The date of the most recent modification of the superstructure, optionally with notes about that modification.

The `NOTE` (p.80) substructure may describe previous changes as well as the most recent, although only the most recent change is described by the `DATE` substructure.

`CREATION_DATE` :=

```
n CREA                                  {1:1}  g7:CREA
  +1 DATE <DateExact>                   {1:1}  g7:DATE-exact
     +2 TIME <Time>                     {0:1}  g7:TIME
```

The date of the initial creation of the superstructure. Because this refers to the initial creation, it should not be modified after the structure is created.

`DATE_VALUE` :=

```
n DATE <DateValue>                      {1:1}  g7:DATE
  +1 TIME <Time>                        {0:1}  g7:TIME
  +1 PHRASE <Text>                      {0:1}  g7:PHRASE
```

A date, optionally with a time and/or a phrase. If there is a `TIME` (p.87), it asserts that the event happened at a specific time on a single day. `TIME` should not be used with `DatePeriod` (p.25) but may be used with other date types.

> *Note* — There is currently no provision for approximate times or time phrases. Time phrases are expected to be added in version 7.1.

`EVENT_DETAIL` :=

```
n <<DATE_VALUE>>                        {0:1}
n <<PLACE_STRUCTURE>>                   {0:1}
n <<ADDRESS_STRUCTURE>>                 {0:1}
n PHON <Special>                        {0:M}  g7:PHON
n EMAIL <Special>                       {0:M}  g7:EMAIL
n FAX <Special>                         {0:M}  g7:FAX
n WWW <Special>                         {0:M}  g7:WWW
n AGNC <Text>                           {0:1}  g7:AGNC
n RELI <Text>                           {0:1}  g7:RELI
n CAUS <Text>                           {0:1}  g7:CAUS
n RESN <List:Enum>                      {0:1}  g7:RESN
n SDATE <DateValue>                     {0:1}  g7:SDATE
  +1 TIME <Time>                        {0:1}  g7:TIME
  +1 PHRASE <Text>                      {0:1}  g7:PHRASE
n <<ASSOCIATION_STRUCTURE>>             {0:M}
n <<NOTE_STRUCTURE>>                    {0:M}
n <<SOURCE_CITATION>>                   {0:M}
n <<MULTIMEDIA_LINK>>                   {0:M}
n UID <Special>                         {0:M}  g7:UID
```

Substructures that may be shared by most individual and family events and attributes.

Note that many of these substructures are limited to 1 per event. Conflicting event information should be represented by placing them in separate event structures (with appropriate source citations) rather than by placing them under the same enclosing event.

`FAMILY_ATTRIBUTE_STRUCTURE` :=

```
[
n NCHI <Integer>                          {1:1}  g7:FAM-NCHI
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>              {0:1}
|
n RESI <Text>                             {1:1}  g7:FAM-RESI
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>              {0:1}
|
n FACT <Text>                             {1:1}  g7:FAM-FACT
  +1 TYPE <Text>                          {1:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>              {0:1}
]
```

Family attributes; see Family Attributes (p.64) for descriptions of each family attribute type.

> *Note* — Family attribute structures vary as follows:
>
> - `FAM.NCHI` (p.79) has an Integer (p.24) payload; others have Text (p.24) payloads
> - `FAM.FACT` (p.72) requires `TYPE` (p.90); it's optional for others

`FAMILY_EVENT_DETAIL` :=

```
n HUSB                                    {0:1}  g7:HUSB
  +1 AGE <Age>                            {1:1}  g7:AGE
     +2 PHRASE <Text>                     {0:1}  g7:PHRASE
n WIFE                                    {0:1}  g7:WIFE
  +1 AGE <Age>                            {1:1}  g7:AGE
     +2 PHRASE <Text>                     {0:1}  g7:PHRASE
n <<EVENT_DETAIL>>                        {0:1}
```

Substructures shared by most family events and attributes.

`FAMILY_EVENT_STRUCTURE` :=

```
[
n ANUL [Y|<NULL>]                         {1:1}  g7:ANUL
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>              {0:1}
```

```
|
n CENS [Y|<NULL>]                          {1:1}  g7:FAM-CENS
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>               {0:1}
|
n DIV [Y|<NULL>]                           {1:1}  g7:DIV
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>               {0:1}
|
n DIVF [Y|<NULL>]                          {1:1}  g7:DIVF
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>               {0:1}
|
n ENGA [Y|<NULL>]                          {1:1}  g7:ENGA
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>               {0:1}
|
n MARB [Y|<NULL>]                          {1:1}  g7:MARB
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>               {0:1}
|
n MARC [Y|<NULL>]                          {1:1}  g7:MARC
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>               {0:1}
|
n MARL [Y|<NULL>]                          {1:1}  g7:MARL
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>               {0:1}
|
n MARR [Y|<NULL>]                          {1:1}  g7:MARR
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>               {0:1}
|
n MARS [Y|<NULL>]                          {1:1}  g7:MARS
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>               {0:1}
|
n EVEN <Text>                              {1:1}  g7:FAM-EVEN
  +1 TYPE <Text>                           {1:1}  g7:TYPE
  +1 <<FAMILY_EVENT_DETAIL>>               {0:1}
]
```

Family events; see Family Events (p.61) for descriptions of each family event type.

An event structure may be used to discuss an event even if the event is not known to have occurred. See Events (p.59) for a discussion of how `DATE` (p.69), `PLAC` (p.82), and the optional `Y` payload indicate whether the structure is asserting the event occurred. See the `NON_EVENT_STRUCTURE` (p.54) for how to state an event did not occur.

> *Note* — Family event structures vary as follows:
>
> - `FAM.EVEN` (p.71) has a Text (p.24) payload; others may have a `Y` payload
> - `FAM.EVEN` requires `TYPE` (p.90); it's optional for others

`IDENTIFIER_STRUCTURE` :=

```
[
n REFN <Special>                          {1:1}  g7:REFN
  +1 TYPE <Text>                          {0:1}  g7:TYPE
|
n UID <Special>                           {1:1}  g7:UID
|
n EXID <Special>                          {1:1}  g7:EXID
  +1 TYPE <Special>                       {0:1}  g7:EXID-TYPE
]
```

> *Deprecation Note* — Having an `EXID` (p.72) without an `EXID.TYPE` (p.90) substructure is deprecated. The meaning of an `EXID` depends on its `EXID.TYPE`. The cardinality of `EXID.TYPE` will be changed to `{1:1}` in version 8.0.

Each of these provides an identifier for a structure or its subject, and each is different in purpose:

- `REFN` (p.83) is a user-generated identifier for a structure.
- `UID` (p.91) is a globally-unique identifier for a structure.
- `EXID` is an identifier maintained by an external authority that applies to the subject of the structure.

`INDIVIDUAL_ATTRIBUTE_STRUCTURE` :=

```
[
n CAST <Text>                             {1:1}  g7:CAST
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>          {0:1}
|
n DSCR <Text>                             {1:1}  g7:DSCR
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>          {0:1}
|
n EDUC <Text>                             {1:1}  g7:EDUC
```

```
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>          {0:1}
|
n IDNO <Special>                          {1:1}  g7:IDNO
  +1 TYPE <Text>                          {1:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>          {0:1}
|
n NATI <Text>                             {1:1}  g7:NATI
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>          {0:1}
|
n NCHI <Integer>                          {1:1}  g7:INDI-NCHI
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>          {0:1}
|
n NMR <Integer>                           {1:1}  g7:NMR
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>          {0:1}
|
n OCCU <Text>                             {1:1}  g7:OCCU
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>          {0:1}
|
n PROP <Text>                             {1:1}  g7:PROP
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>          {0:1}
|
n RELI <Text>                             {1:1}  g7:INDI-RELI
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>          {0:1}
|
n RESI <Text>                             {1:1}  g7:INDI-RESI
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>          {0:1}
|
n SSN <Special>                           {1:1}  g7:SSN
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>          {0:1}
|
n TITL <Text>                             {1:1}  g7:INDI-TITL
  +1 TYPE <Text>                          {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>          {0:1}
|
n FACT <Text>                             {1:1}  g7:INDI-FACT
  +1 TYPE <Text>                          {1:1}  g7:TYPE
```

```
  +1 <<INDIVIDUAL_EVENT_DETAIL>>            {0:1}
]
```

Individual attributes; see Individual Attributes (p.62) for descriptions of each individual attribute type..

> *Note* — Individual attribute structures vary as follows:
>
> - `INDI.NCHI` (p.79) and `NMR` (p.79) have Integer (p.24) payloads; `IDNO` (p.75) and `SSN` (p.86) have Special (p.31) payloads; others have Text (p.24) payloads
> - `INDI.FACT` (p.73) and `IDNO` require `TYPE` (p.90); it's optional for others

`INDIVIDUAL_EVENT_DETAIL` :=

```
n <<EVENT_DETAIL>>                          {1:1}
n AGE <Age>                                 {0:1}  g7:AGE
  +1 PHRASE <Text>                          {0:1}  g7:PHRASE
```

Substructures shared by most individual events and attributes.

`INDIVIDUAL_EVENT_STRUCTURE` :=

```
[
n ADOP [Y|<NULL>]                           {1:1}  g7:ADOP
  +1 TYPE <Text>                            {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>            {0:1}
  +1 FAMC @<XREF:FAM>@                      {0:1}  g7:ADOP-FAMC
     +2 ADOP <Enum>                         {0:1}  g7:FAMC-ADOP
        +3 PHRASE <Text>                    {0:1}  g7:PHRASE
|
n BAPM [Y|<NULL>]                           {1:1}  g7:BAPM
  +1 TYPE <Text>                            {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>            {0:1}
|
n BARM [Y|<NULL>]                           {1:1}  g7:BARM
  +1 TYPE <Text>                            {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>            {0:1}
|
n BASM [Y|<NULL>]                           {1:1}  g7:BASM
  +1 TYPE <Text>                            {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>            {0:1}
|
n BIRT [Y|<NULL>]                           {1:1}  g7:BIRT
  +1 TYPE <Text>                            {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>            {0:1}
  +1 FAMC @<XREF:FAM>@                      {0:1}  g7:FAMC
|
```

```
n BLES [Y|<NULL>]                          {1:1}  g7:BLES
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n BURI [Y|<NULL>]                          {1:1}  g7:BURI
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n CENS [Y|<NULL>]                          {1:1}  g7:INDI-CENS
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n CHR [Y|<NULL>]                           {1:1}  g7:CHR
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
  +1 FAMC @<XREF:FAM>@                     {0:1}  g7:FAMC
|
n CHRA [Y|<NULL>]                          {1:1}  g7:CHRA
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n CONF [Y|<NULL>]                          {1:1}  g7:CONF
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n CREM [Y|<NULL>]                          {1:1}  g7:CREM
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n DEAT [Y|<NULL>]                          {1:1}  g7:DEAT
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n EMIG [Y|<NULL>]                          {1:1}  g7:EMIG
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n FCOM [Y|<NULL>]                          {1:1}  g7:FCOM
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n GRAD [Y|<NULL>]                          {1:1}  g7:GRAD
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
```

```
n IMMI [Y|<NULL>]                          {1:1}  g7:IMMI
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n NATU [Y|<NULL>]                          {1:1}  g7:NATU
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n ORDN [Y|<NULL>]                          {1:1}  g7:ORDN
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n PROB [Y|<NULL>]                          {1:1}  g7:PROB
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n RETI [Y|<NULL>]                          {1:1}  g7:RETI
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n WILL [Y|<NULL>]                          {1:1}  g7:WILL
  +1 TYPE <Text>                           {0:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
|
n EVEN <Text>                              {1:1}  g7:INDI-EVEN
  +1 TYPE <Text>                           {1:1}  g7:TYPE
  +1 <<INDIVIDUAL_EVENT_DETAIL>>           {0:1}
]
```

Individual events; see Individual Events (p.60) for descriptions of each individual event type.

An event structure may be used to discuss an event even if the event is not known to have occurred. See Events (p.59) for a discussion of how `DATE` (p.69), `PLAC` (p.82), and the optional `Y` payload indicate whether the structure is asserting the event occurred. See the `NON_EVENT_STRUCTURE` (p.54) for how to state an event did not occur.

> *Note* — Individual event structures vary as follows:
>
> - `INDI.EVEN` (p.71) has a Text (p.24) payload; others may have a `Y` payload
> - `INDI.EVEN` requires `TYPE` (p.90); it's optional for others
> - `BIRT` (p.67) and `CHR` (p.68) may have a `FAMC` (p.73) with no substructures; `ADOP` (p.65) may have a `FAMC` with an optional `ADOP` substructure; others may not have a `FAMC` substructure. The `FAMC` substructure can itself have substructures for source citations and notes related to the child's relationship to parents, and is the recommended place to store such information.

LDS_INDIVIDUAL_ORDINANCE :=

```
[
n BAPL                                    {1:1}  g7:BAPL
  +1 <<LDS_ORDINANCE_DETAIL>>             {0:1}
|
n CONL                                    {1:1}  g7:CONL
  +1 <<LDS_ORDINANCE_DETAIL>>             {0:1}
|
n ENDL                                    {1:1}  g7:ENDL
  +1 <<LDS_ORDINANCE_DETAIL>>             {0:1}
|
n INIL                                    {1:1}  g7:INIL
  +1 <<LDS_ORDINANCE_DETAIL>>             {0:1}
|
n SLGC                                    {1:1}  g7:SLGC
  +1 <<LDS_ORDINANCE_DETAIL>>             {0:1}
  +1 FAMC @<XREF:FAM>@                    {1:1}  g7:FAMC
]
```

Ordinances performed by members of The Church of Jesus Christ of Latter-day Saints; see Latter-day Saint Ordinances (p.64) for descriptions of each ordinance type.

LDS_ORDINANCE_DETAIL :=

```
n <<DATE_VALUE>>                         {0:1}
n TEMP <Text>                            {0:1}  g7:TEMP
n <<PLACE_STRUCTURE>>                    {0:1}
n STAT <Enum>                            {0:1}  g7:ord-STAT
  +1 DATE <DateExact>                    {1:1}  g7:DATE-exact
     +2 TIME <Time>                      {0:1}  g7:TIME
n <<NOTE_STRUCTURE>>                     {0:M}
n <<SOURCE_CITATION>>                    {0:M}
```

Dates for these ordinances should be in the default (`GREGORIAN` (p.103)) calendar and be 1830 or later. These ordinances can be performed posthumously by proxy, and the date may reflect that posthumous date.

LDS_SPOUSE_SEALING :=

```
n SLGS                                    {1:1}  g7:SLGS
  +1 <<LDS_ORDINANCE_DETAIL>>             {0:1}
```

Ordinances performed by members of The Church of Jesus Christ of Latter-day Saints; see Latter-day Saint Ordinances (p.64) for descriptions of each ordinance type.

MULTIMEDIA_LINK :=

```
n OBJE @<XREF:OBJE>@                          {1:1}  g7:OBJE
  +1 CROP                                     {0:1}  g7:CROP
     +2 TOP <Integer>                         {0:1}  g7:TOP
     +2 LEFT <Integer>                        {0:1}  g7:LEFT
     +2 HEIGHT <Integer>                      {0:1}  g7:HEIGHT
     +2 WIDTH <Integer>                       {0:1}  g7:WIDTH
  +1 TITL <Text>                              {0:1}  g7:TITL
```

Links the superstructure to the `MULTIMEDIA_RECORD` (p.40) with the given pointer.

The optional `CROP` (p.69) substructure indicates that a subregion of an image represents or applies to the superstructure.

The optional `TITL` (p.87) substructure supersedes any `OBJE.FILE.TITL` substructures included in the `MULTIMEDIA_RECORD`.

NON_EVENT_STRUCTURE :=

```
n NO <Enum>                                   {1:1}  g7:NO
  +1 DATE <DatePeriod>                        {0:1}  g7:NO-DATE
     +2 PHRASE <Text>                         {0:1}  g7:PHRASE
  +1 <<NOTE_STRUCTURE>>                       {0:M}
  +1 <<SOURCE_CITATION>>                      {0:M}
```

Indicates that a specific type of event, given in the payload, did not happen within a given date period (or never happened if there is no `DATE` (p.69) substructure).

Substructures may provide discussion about the non-occurrence of the event but must not limit the meaning of what did not occur. No substructure other than `DATE` may restrict the breadth of that negative assertion.

> *Example* — `1 NO MARR` means “no marriage occurred”

> *Example* —
>
> ```
> 1 NO MARR
> 2 DATE TO 24 MAR 1880
> ```
>
> means “no marriage had occurred as of March 24th, 1880”

NOTE_STRUCTURE :=

```
[
n NOTE <Text>                                {1:1}  g7:NOTE
  +1 MIME <MediaType>                        {0:1}  g7:MIME
  +1 LANG <Language>                         {0:1}  g7:LANG
  +1 TRAN <Text>                             {0:M}  g7:NOTE-TRAN
     +2 MIME <MediaType>                     {0:1}  g7:MIME
     +2 LANG <Language>                      {0:1}  g7:LANG
  +1 <<SOURCE_CITATION>>                     {0:M}
|
n SNOTE @<XREF:SNOTE>@                       {1:1}  g7:SNOTE
]
```

A catch-all location for information that does not fully fit within other structures. It may include research notes, additional context, alternative interpretations, reasoning, and so forth.

Each NOTE.TRAN (p.89) must have either a MIME (p.78) or LANG (p.75) substructure, and may have both.

See SHARED_NOTE_RECORD (p.41) for advice on choosing between NOTE (p.80) and SNOTE (p.85).

A NOTE_STRUCTURE (p.55) can contain a SOURCE_CITATION (p.58), which in turn can contain a NOTE_STRUCTURE, allowing potentially unbounded nesting of structures. Because each dataset is finite, this nesting is also guaranteed to be finite.

PERSONAL_NAME_PIECES :=

```
n NPFX <Text>                                {0:M}  g7:NPFX
n GIVN <Text>                                {0:M}  g7:GIVN
n NICK <Text>                                {0:M}  g7:NICK
n SPFX <Text>                                {0:M}  g7:SPFX
n SURN <Text>                                {0:M}  g7:SURN
n NSFX <Text>                                {0:M}  g7:NSFX
```

Optional isolated name parts; see PERSONAL_NAME_STRUCTURE for more details.

> *Example* — “Lt. Cmndr. Joseph Allen jr.” might be presented as
>
> ```
> 1 NAME Lt. Cmndr. Joseph /Allen/ jr.
> 2 NPFX Lt. Cmndr.
> 2 GIVN Joseph
> 2 SURN Allen
> 2 NSFX jr.
> ```

This specification does not define how the meaning of multiple parts with the same tag differs from the meaning of a single part with a concatenated larger payload. However, some applications allow the user to chose whether to combine or split name parts, meaning the tag quantity should be treated as expressing at least a user preference. Even when multiple `SURN` (p.86) tags are used, the `PersonalName` (p.30) data type identifies a single surname substring between its slashes.

`PERSONAL_NAME_STRUCTURE` :=

```
n NAME <PersonalName>                    {1:1}  g7:INDI-NAME
  +1 TYPE <Enum>                         {0:1}  g7:NAME-TYPE
     +2 PHRASE <Text>                    {0:1}  g7:PHRASE
  +1 <<PERSONAL_NAME_PIECES>>            {0:1}
  +1 TRAN <PersonalName>                 {0:M}  g7:NAME-TRAN
     +2 LANG <Language>                  {1:1}  g7:LANG
     +2 <<PERSONAL_NAME_PIECES>>         {0:1}
  +1 <<NOTE_STRUCTURE>>                  {0:M}
  +1 <<SOURCE_CITATION>>                 {0:M}
```

Names of individuals are represented in the manner the name is normally spoken, with the family name, surname, or nearest cultural parallel thereunto separated by slashes (U+002F `/`). Based on the dynamic nature or unknown compositions of naming conventions, it is difficult to provide a more detailed name piece structure to handle every case. The `PERSONAL_NAME_PIECES` (p.55) are provided optionally for systems that cannot operate effectively with less structured information. The Personal Name payload shall be seen as the primary name representation, with name pieces as optional auxiliary information; in particular it is recommended that all name parts in `PERSONAL_NAME_PIECES` appear within the `PersonalName` payload in some form, possibly adjusted for gender-specific suffixes or the like. It is permitted for the payload to contain information not present in any name piece substructure.

The name may be translated or transliterated into different languages or scripts using the `TRAN` (p.87) substructure. It is recommended, but not required, that if the name pieces are used, the same pieces are used in each translation and transliteration.

A `TYPE` (p.90) is used to specify the particular variation that this name is. For example; it could indicate that this name is a name taken at immigration or that it could be an ‘also known as’ name. See `g7:enumset-NAME-TYPE` (p.98) for more details.

> *Note* — Alternative approaches to representing names are being considered for future versions of this specification.

PLACE_STRUCTURE :=

```
n PLAC <List:Text>                          {1:1}  g7:PLAC
  +1 FORM <List:Text>                       {0:1}  g7:PLAC-FORM
  +1 LANG <Language>                        {0:1}  g7:LANG
  +1 TRAN <List:Text>                       {0:M}  g7:PLAC-TRAN
     +2 LANG <Language>                     {1:1}  g7:LANG
  +1 MAP                                    {0:1}  g7:MAP
     +2 LATI <Special>                      {1:1}  g7:LATI
     +2 LONG <Special>                      {1:1}  g7:LONG
  +1 EXID <Special>                         {0:M}  g7:EXID
     +2 TYPE <Special>                      {0:1}  g7:EXID-TYPE
  +1 <<NOTE_STRUCTURE>>                     {0:M}
```

> *Deprecation Note* — Having an `EXID` (p.72) without an `EXID.TYPE` (p.90) substructure is deprecated. The meaning of an `EXID` depends on its `EXID.TYPE`. The cardinality of `EXID.TYPE` will be changed to `{1:1}` in version 8.0.

A place, which can be represented in several ways:

- The payload contains a comma-separated list of region names, ordered from smallest to largest. The specific meaning of each element is given by the `FORM` (p.74) substructure, or in the `HEAD.PLAC.FORM` (p.74) if there is no `FORM` substructure. If neither `FORM` exists, the meaning of the elements are not defined in this specification beyond being names of jurisdictions of some kind, ordered from smallest to largest.

  > Some applications and users have defaulted to assuming a *Note* — `FORM` of "City, County, State, Country", and some applications even ignore any `FORM` substructures and treat payloads with a smaller number of elements as if they had additional blank elements at the end.

  Elements should be left blank if they are unknown, do not apply to the location, or are too specific for the region in question.

  > *Example* — A record describing births throughout Oneida county could be recorded as
  >
  > ```
  > 0 @S1@ SOUR
  > 1 DATA
  > 2 EVEN BIRT
  > 3 PLAC , Oneida, Idaho, USA
  > 4 FORM City, County, State, Country
  > ```

- The payload may be translated or transliterated into different languages or scripts using the `TRAN` (p.87) substructure. It should use the same `FORM` as the payload.
- Global coordinates may be presented in the `MAP` (p.77) substructure

> *Note* — This specification does not support places where a region name contains a comma. An alternative system for representing locations is likely to be added in a later version.

SOURCE_CITATION :=

```
n SOUR @<XREF:SOUR>@                    {1:1}  g7:SOUR
  +1 PAGE <Text>                        {0:1}  g7:PAGE
  +1 DATA                               {0:1}  g7:SOUR-DATA
     +2 <<DATE_VALUE>>                  {0:1}
     +2 TEXT <Text>                     {0:M}  g7:TEXT
        +3 MIME <MediaType>             {0:1}  g7:MIME
        +3 LANG <Language>              {0:1}  g7:LANG
  +1 EVEN <Enum>                        {0:1}  g7:SOUR-EVEN
     +2 PHRASE <Text>                   {0:1}  g7:PHRASE
     +2 ROLE <Enum>                     {0:1}  g7:ROLE
        +3 PHRASE <Text>                {0:1}  g7:PHRASE
  +1 QUAY <Enum>                        {0:1}  g7:QUAY
  +1 <<MULTIMEDIA_LINK>>                {0:M}
  +1 <<NOTE_STRUCTURE>>                 {0:M}
```

A citation indicating that the pointed-to source record supports the claims made in the superstructure. Substructures provide additional information about how that source applies to the subject of the citation's superstructure:

- PAGE (p.80): where in the source the relevant material can be found.
- DATA (p.69): the relevant data from the source.
- EVEN: what event the relevant material was recording.
- QUAY (p.83): an estimation of the reliability of the source in regard to these claims.
- MULTIMEDIA_LINK (p.54): digital copies of the cited part of the source

It is recommended that every SOURCE_CITATION (p.58) point to a SOURCE_RECORD (p.42). However, a voidPtr (p.11) can be used with the citation text in a PAGE substructure. The PAGE is defined to express a "specific location within the information referenced;" with a voidPtr there is no information referenced, so the PAGE may describe the entire source.

A SOURCE_CITATION can contain a NOTE_STRUCTURE (p.55), which in turn can contain a SOURCE_CITATION, allowing potentially unbounded nesting of structures. Because each dataset is finite, this nesting is also guaranteed to be finite.

SOURCE_REPOSITORY_CITATION :=

```
n REPO @<XREF:REPO>@                    {1:1}  g7:REPO
  +1 <<NOTE_STRUCTURE>>                 {0:M}
  +1 CALN <Special>                     {0:M}  g7:CALN
     +2 MEDI <Enum>                     {0:1}  g7:MEDI
        +3 PHRASE <Text>                {0:1}  g7:PHRASE
```

This structure is used within a source record to point to a name and address record of the holder of the source document. Formal and informal repository name and addresses are stored in the `REPOSITORY_RECORD` (p.40). More formal repositories, such as the Family History Library, should show a call number of the source at that repository. The call number of that source should be recorded using a `CALN` (p.67) substructure.

3.3. Structure Meaning

3.3.1. Events

As a general rule, events are things that happen on a specific date. Use the `dateRange` (p.25) form to indicate that an event took place at some time between 2 dates. In most cases, a `DatePeriod` is inappropriate for an event; if the subject of your recording occurred over a period of time, then it is probably not an event, but rather an attribute.

Event structures can be used to record notes about an event without asserting the event actually occurred. An event structure asserts the event did occur if any of the following are true:

- There is a `DATE` (p.69) substructure

Example —

```
1 DEAT
2 DATE 2 OCT 1937
```

> *Note* — Version 5.4 (1995) introduced the "event did occur" meaning of event.`DATE`, so it is now well-established in applications and files. However, it is common for users to enter a date range with no end without intending to indicate that the event occurred. For example, pre 7.0 files sometimes used
>
> ```
> 1 NATU
> 2 DATE AFT 1800
> ```
>
> to mean what 7.0 encodes as
>
> ```
> 1 NOT NATU
> 2 DATE TO 1800
> ```
>
> without intending to imply that `NATU` (p.79) ever did actually occur. Because this is a "sometimes used" rather than a "formally means" situation, it is likely that data using 5.x "after meaning not before" *de facto* pattern will be transferred as-is into 7.0 and persist in files for the foreseeable future.

- There is a `PLAC` (p.82) substructure

Example —

```
1 DEAT
2 PLAC Cove, Cache, Utah
```

- The event has a payload. A special payload `Y` can be used with some event types to indicate that the event is known to have occurred without providing any additional information about it.

Example —

```
1 DEAT Y
```

If none of the above are true, the structure should be seen as a place for inconclusive research notes about the possibility of the event. An assertion that an event did not occur should be encoded using the `NO` (p.79) structure.

3.3.1.1. Individual Events

Tag	Name URI	Description
`ADOP`	adoption `g7:ADOP` (p.65)	Creation of a legally approved child-parent relationship that does not exist biologically.
`BAPM`	baptism `g7:BAPM` (p.67)	Baptism, performed in infancy or later. (See also `BAPL` (p.64) and `CHR`.)
`BARM`	Bar Mitzvah `g7:BARM` (p.67)	The ceremonial event held when a Jewish boy reaches age 13.
`BASM`	Bas Mitzvah `g7:BASM` (p.67)	The ceremonial event held when a Jewish girl reaches age 13, also known as “Bat Mitzvah.”
`BIRT`	birth `g7:BIRT` (p.67)	Entering into life.
`BLES`	blessing `g7:BLES` (p.67)	Bestowing divine care or intercession. Sometimes given in connection with a naming ceremony.
`BURI`	depositing remains `g7:BURI` (p.67)	Depositing the mortal remains of a deceased person.
`CENS`	census `g7:INDI-CENS` (p.67)	Periodic count of the population for a designated locality, such as a national or state census.
`CHR`	christening `g7:CHR` (p.68)	Baptism or naming events for a child.
`CHRA`	adult christening `g7:CHRA` (p.68)	Baptism or naming events for an adult person.
`CONF`	confirmation `g7:CONF` (p.68)	Conferring full church membership.